

Applying optimality conditions to projection

Obtaining a geometric characterization of projection

Recall: $\min: f(x)$ $\hat{x}$ optimal iff $0 \in \partial f(\hat{x}) + N_{\mathcal{X}}(\hat{x})$.
st: $x \in \mathcal{X}$



$$\min: \frac{1}{2} \|x - y\|_2^2$$
$$\text{st: } x \in \mathcal{X}$$

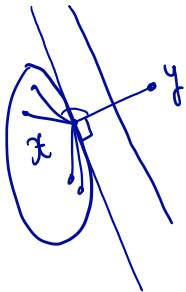
$\hat{x}$ is an opt. sol'n iff

$$0 \in \partial f(\hat{x}) + N_{\mathcal{X}}(\hat{x})$$

$$\Rightarrow 0 \in \hat{x} - y + N_{\mathcal{X}}(\hat{x}) \Rightarrow \underline{y - \hat{x}} \in N_{\mathcal{X}}(\hat{x})$$

$$\langle y - \hat{x}, x - \hat{x} \rangle \leq 0 \quad \forall x \in \mathcal{X}$$

Separating hyperplane



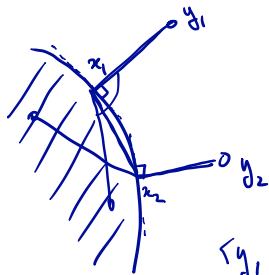
given a point $y \notin X$ how do we find a separating hyperplane?

Exercise: This picture shows us that the normal of a separating hyperplane is $(y - \hat{x})$. Show that this in fact works.

$$H_{s,b} = \{x : s^T x = b\}, \quad H_{s,b}^+, H_{s,b}^-$$

Hint: $s = (y - \hat{x})$, where $\hat{x}$ is the projection of y onto X .

Projection onto a convex set is a contraction



$$\|y_1 - y_2\|_2 \geq \|x_1 - x_2\|_2$$

Recall:

$$\langle y_1 - x_1, x_2 - x_1 \rangle \leq 0$$

$$\langle y_2 - x_2, x_1 - x_2 \rangle \leq 0$$

$$\langle y_1 - x_1, x_2 - x_1 \rangle - \langle y_2 - x_2, x_2 - x_1 \rangle \leq 0$$

$$\langle y_1 - y_2 - (x_1 - x_2), x_1 - x_2 \rangle \geq 0$$

$$\langle y_1 - y_2, x_1 - x_2 \rangle - \langle x_1 - x_2, x_1 - x_2 \rangle \geq 0$$

$$\|y_1 - y_2\| \cdot \|x_1 - x_2\| \geq \langle y_1 - y_2, x_1 - x_2 \rangle \geq \|x_1 - x_2\|^2 \implies \|y_1 - y_2\| \geq \|x_1 - x_2\|.$$

For non-convex
no longer true:



$$\langle u, v \rangle \leq \|u\| \cdot \|v\|$$